

Figure 1 assets: global-time slot labels

Predicted latent-state slots

$$\hat{z}_{t+1}^1, \quad \hat{z}_{t+2}^1, \quad \dots, \quad \hat{z}_{t+H}^1$$

Predicted action slots

$$\hat{a}_t^1, \quad \hat{a}_{t+1}^1, \quad \dots, \quad \hat{a}_{t+H-1}^1$$

Real environment-state slots

$$s_t, \quad s_{t+1}, \quad s_{t+2}, \quad \dots, \quad s_{t+H}$$

Encoded state-feedback slots

$$z_{t+1}, \quad z_{t+2}, \quad \dots, \quad z_{t+H}$$

Committed actions from the preceding chunk

$$a_t^{\text{prev}}, \quad a_{t+1}^{\text{prev}}, \quad \dots, \quad a_{t+H-1}^{\text{prev}}$$

Aligned cross-chunk action overlap

$$i \in \mathcal{I}_t^A, \quad a_{t+i}^{\text{prev}}, \quad i = 0, \dots, H-1$$

Figure 1 assets: stage-wise FBFM formulas

Execution transition and encoded feedback

$$s_{t+i} \sim P(\cdot \mid s_{t+i-1}, a_{t+i-1}^{\text{prev}}),$$
$$z_{t+i} = E(\mathcal{O}(s_{t+i})), \quad (i, z_{t+i}) \in \mathcal{F}_{t,k}$$

State-flow correction

$$\mathbf{e}_{t,k}^Z = P_Z \mathbf{W}_{t,k}^Z \left(\mathbf{Y}_{t,k}^Z - \hat{\mathbf{Z}}_{t,k}^1 \right),$$
$$v_{\text{FBFM}}^Z = v_{\theta_Z}^Z + \lambda_{\tau_k^Z}^Z (\mathbf{J}_{t,k}^Z)^\top \mathbf{e}_{t,k}^Z$$

Action-flow correction

$$\mathbf{e}_{t,k}^A = P_A \mathbf{W}_t^A \left(\mathbf{Y}_t^A - \hat{\mathbf{A}}_{t,k}^1 \right),$$
$$v_{\text{FBFM}}^A = v_{\theta_A}^A + \lambda_{\tau_k^A}^A (\mathbf{J}_{t,k}^A)^\top \mathbf{e}_{t,k}^A$$

Feedback-refreshed state context

$$\check{\mathbf{Z}}_{t,k} = (\mathbf{I}_{D_Z} - \mathbf{W}_{t,k}^Z) \hat{\mathbf{Z}}_t + \mathbf{W}_{t,k}^Z \mathbf{Y}_{t,k}^Z,$$
$$\mathcal{C}_{t,k}^Z = \Phi_Z(\check{\mathbf{Z}}_{t,k}, \mathcal{H}_t)$$

Figure 3 assets: joint-generation FBFM

Joint state–action variable

$$\mathbf{X}_t = \begin{bmatrix} \mathbf{Z}_t \\ \mathbf{A}_t \end{bmatrix}, \quad v_\theta^X : \mathbf{X}_t^\tau \mapsto \frac{d\mathbf{X}_t^\tau}{d\tau}$$

Joint feedback target, mask, and preconditioner

$$\mathbf{Y}_{t,k}^X = \begin{bmatrix} \mathbf{Y}_{t,k}^Z \\ \mathbf{Y}_t^A \end{bmatrix}, \quad \mathbf{W}_{t,k}^X = \begin{bmatrix} \mathbf{W}_{t,k}^Z & \mathbf{0} \\ \mathbf{0} & \mathbf{W}_t^A \end{bmatrix},$$

$$\mathbf{P}^X = \text{blkdiag}(P_Z \mathbf{I}_{D_Z}, P_A \mathbf{I}_{D_A})$$

Joint guided velocity field

$$\mathbf{e}_{t,k}^X = \mathbf{P}^X \mathbf{W}_{t,k}^X \left(\mathbf{Y}_{t,k}^X - \hat{\mathbf{X}}_{t,k}^1 \right),$$

$$v_{\text{FBFM}}^X = v_\theta^X + \lambda_{\tau_k^X}^X \left(\mathbf{J}_{t,k}^X \right)^\top \mathbf{e}_{t,k}^X$$

Direct state-to-action correction

$$\mathbf{e}_{t,k}^A = \mathbf{0} \quad \implies \quad \mathbf{g}_{t,k}^A = \mathbf{J}_{ZA}^\top \mathbf{e}_{t,k}^Z = \left(\frac{\partial \hat{\mathbf{Z}}_{t,k}^1}{\partial \mathbf{A}_t^{\tau_k^X}} \right)^\top \mathbf{e}_{t,k}^Z$$

Shared replacement formulas and compact labels

Expanded one-line FBFM flow integral

$$\mathbf{Q}_t^1 = \mathbf{Q}_t^0 + \int_0^1 \left[v_{\theta_Q}^Q \left(\mathbf{Q}_t^\tau, \tau; \mathcal{K}_t^Q(\tau) \right) + \lambda_\tau^Q \left(\mathbf{J}_t^Q(\tau) \right)^\top \mathbf{P}^Q \mathbf{W}_t^Q(\tau) \left(\mathbf{Y}_t^Q(\tau) - \hat{\mathbf{Q}}_t^1(\tau) \right) \right] d\tau, \quad Q \in \{Z, A, X\}$$

Unified clean-endpoint and guidance chain

$$\begin{aligned} \hat{\mathbf{Q}}_{t,k}^1 &= \mathbf{Q}_t^{\tau_k^Q} + (1 - \tau_k^Q) v_{\theta_Q}^Q \left(\mathbf{Q}_t^{\tau_k^Q}, \tau_k^Q; \mathcal{K}_{t,k}^Q \right), \\ \mathbf{e}_{t,k}^Q &= \mathbf{P}^Q \mathbf{W}_{t,k}^Q \left(\mathbf{Y}_{t,k}^Q - \hat{\mathbf{Q}}_{t,k}^1 \right), \\ v_{\text{FBFM}}^Q &= v_{\theta_Q}^Q + \lambda_{\tau_k^Q}^Q \left(\mathbf{J}_{t,k}^Q \right)^\top \mathbf{e}_{t,k}^Q \end{aligned}$$

Measurement and generalized lifting

$$\begin{aligned} \mathbf{m}_{t,k}^Q &= h_Q(\mathbf{Q}_t^1) + \boldsymbol{\eta}, \\ \mathbf{Y}_{t,k}^Q &:= h_Q^\dagger(\mathbf{m}_{t,k}^Q), \quad Q \in \{Z, A, X\} \end{aligned}$$

Stage-wise factorization

$$p_\theta(\mathbf{Z}_t, \mathbf{A}_t \mid \mathcal{H}_t) = p_{\theta_Z}(\mathbf{Z}_t \mid \mathcal{H}_t) p_{\theta_A}(\mathbf{A}_t \mid \mathbf{Z}_t, \mathcal{H}_t)$$

Compact diagram labels

Compact text labels

State Flow v_{FBFM}^Z

Action Flow v_{FBFM}^A

Joint State–Action Flow v_{FBFM}^X

Dynamic State Feedback $\mathcal{F}_{t,k}$

Committed Action Overlap $\mathcal{I}_t^A$

Frozen WAM $\theta, \theta_Z, \theta_A$ fixed